(Write your Enrollment Number) Enrollment No.

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name : B. Tech(AIML)) (Semester : IV)

(May, 2024)

Subject Code: BAM-202 **Subject:** Machine Learning

Time :3 Hours **Maximum Marks :60**

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)	

(a) Suppose a hospital builds a cancer detection classification algorithm to classify whether a patient has cancer based on numerous lab values. Here, the class 'Yes' indicates that a person has cancer and the class 'No' indicates that they do not have cancer. Assume that the class 'Yes' is the class of interest. Here is the confusion matrix for the cancer detection model.

		Predicted Class	
		Yes	No
Actual Class	Yes	450	160
	No	70	320

Under some strong assumptions, one can imagine that a delayed cancer diagnosis may lead to worse patient prognosis (more severe disease). In contrast a false alarm may lead to additional testing, cost and potentially anxiety. Weighing these two outcomes, one may conclude that in this case the cost of false negatives is much higher than the cost of false positives.

Compute the misclassification cost for the above model, assuming that the cost of misclassifying an actual cancer patient is Rs1,000 and that for a non-cancer patient is Rs 50.

(b) You plotted the female height population distribution. The distribution is normal with mean μ=66 inches and standard deviation σ=3.5. You have randomly picked a female individual from the population. What is the probability that the individual is taller than 62.5 inches but shorter than 69.5 inches?

(c) With reference to logistic regression, if you will travel along with ROC curve, how cutoff value will change? Explain?

(d) A hypothetical split in a tree has resulted in two children, one with 10 observations (8 being "Hi Risk" and 2 being "Low Risk") and the other with 20 observations (15 being "Low Risk" and 5 being "Hi Risk"). What is the goodness of this hypothetical split? Use accuracy as the metric for goodness of each node.

(e) Discuss the impact of choosing different initial centroid positions on the convergence and quality of k-means clustering.

(f) Explain the curse of dimensionality and its implications for machine learning tasks. How do dimensionality reduction techniques mitigate this issue?

(g) A boosted strong learner L is formed by three weak learners, L1 , L2 , and L3 . Their weights are 1, 0.4, and 1.2, respectively. For a particular point, L1 and L2

predict that its label is positive, and L3 predicts that it's negative. What is the final prediction the learner L makes on this point?

(h)What is the difference between hard and soft voting classifiers?

UNIT-I

Q2	(a)Discuss common statistical measures used for data exploration, such as mean, median, standard deviation, and correlation coefficient. How are they used to understand the characteristics of a dataset? (b) Discuss the trade-offs between feature extraction and feature reduction techniques. How do they impact model interpretability and computational efficiency?	(10)
Q3	(a)Describe the concept of feature extraction versus feature selection in dimensionality reduction. Provide examples of techniques for each approach. (b) Describe K-Fold Cross Validation and its advantages over a simple train-test split. How does it address the limitations of a single train-test split?	(10)

UNIT-II

Q4	A dataset of patients who have tested positive or negative for COVID-19 follows. Their symptoms are cough (C), fever (F), difficulty breathing (B), and tiredness (T). Tabel-1	(10)

	Cough (C)	Fever (F)	Difficulty breathing (B)	Tiredness (T)	Diagnosis
Patient 1	no	yes	yes	yes	Sick
Patient 2	yes	yes	no	yes	Sick
Patient 3	yes	no	yes	yes	Sick
Patient 4	yes	yes	yes	no	Sick
Patient 5	yes	no	no	yes	Healthy
Patient 6	no	yes	yes	no	Healthy
Patient 7	no	yes	no	no	Healthy
Patient 8	no	no	no	yes	Healthy

Build a naive Bayes model that predicts the diagnosis from the symptoms. Use the naive Bayes algorithm to find the following probabilities:

Note: For the following questions, the symptoms that are NO mentioned are completely unknown to us. For example, if we know that the patient has a cough, but nothing is said about their fever, it does not mean the patient doesn't have a fever.

a. The probability that a patient is sick given that the patient has a cough

b. The probability that a patient is sick given that the patient is not tired

	c. The probability that a patient is sick given that the patient has a cough and a fever d. The probability that a patient is sick given that the patient has a cough and a fever, but no difficulty breathing.		
Q5	A dataset representing patients tested for COVID-19, indicating whether they tested positive or negative, is derived from the table provided above(Table-1). Their symptoms are cough (C), fever (F), difficulty breathing (B), and tiredness (T). Using the Gini impurity index, build a decision tree of height 1 that classifies (Diagnosis) this data. What is the accuracy of this classifier on the dataset?	(10)	
	UNIT-III		
Q6	(a)Given the following set of points: P1(2,3), P2(2,4), P3(3,2), P4(3,3), P5(3,4), P6(4,2), P7(4,3) Compute full iterations of k-means clustering for K=2 with initial clusters Centroid μ1 = (2, 3) and μ2 = (3, 4). Make sure to write down the necessary distances (Euclidean distance), explain the steps you follow, and to describe the resulting clusters (centroid and points) at the end of each iteration. (b) Discuss the challenges associated with unsupervised learning compared to supervised learning. How does the absence of labeled data affect model training and evaluation?	(7+3)	
Q7	(a)Describe the purpose of dimensionality reduction techniques in machine learning. How do they help in improving model performance and interpretability? (b) Explain the concept of explained variance ratio in PCA. How can it be used to determine the number of principal components to retain?	(10)	
	UNIT-IV		
Q8	(a)Define Random Forests and explain how they differ from traditional decision trees. What is the role of bagging in building Random Forests? (b)) Suppose you are using Ridge Regression and you notice that the training error and the validation error are almost equal and fairly high. Would you say that the model suffers from high bias or high variance? Should you increase the regularization hyperparameter α or reduce it? Explain?	(6+4)	
Q9	(a)Discuss the advantages of Random Forests over single decision trees in terms of predictive performance and model interpretability. (b) If you have trained five different models on the exact same training data, and they all achieve 95% precision, is there any chance that you can combine these models to get better results? If so, how? If not, why?	(10)	